SURAKSHA VAULT

UNLOCK TRUST, NOT YOUR DATA.

केनरा बैंक
Canara Bank

Team - Suraksha kavach

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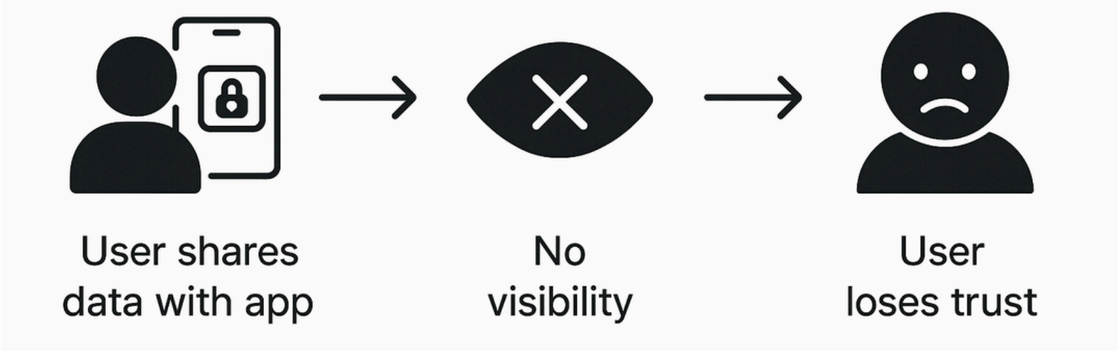
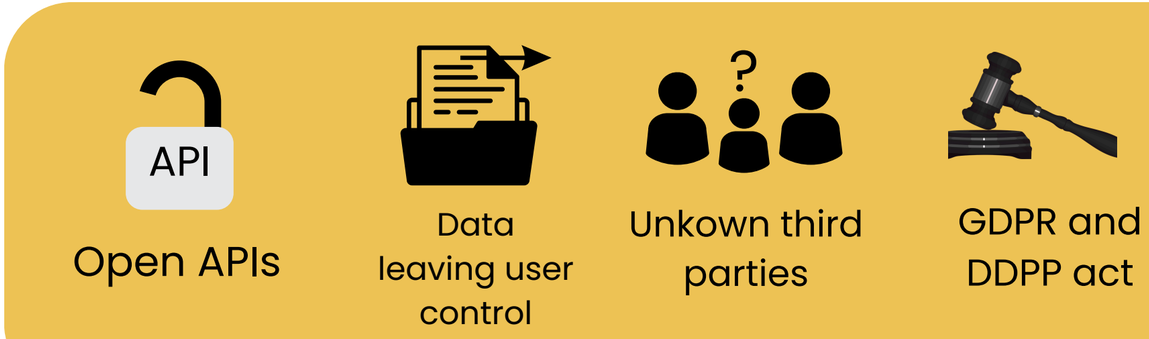
WHY FINTECH NEEDS SMARTER PRIVACY CONTROLS?

PROBLEM STATEMENT

How can fintech platforms enable secure, privacy-preserving, and regulation-compliant data sharing with third-party services—while minimizing data exposure, enforcing usage boundaries, and maintaining end-user trust and control across a distributed data ecosystem?

- Fintech apps handle highly sensitive data (PII, income, transactions).
- Third-party integrations (Open Banking, APIs) increase data exposure.
- Once shared, data control is lost – can't stop misuse or reselling.
- Users lack visibility and revocation options after giving consent.
- Rising regulatory pressure (GDPR, DPDP) demands transparency + control.





MEET SURAKSHA VAULT

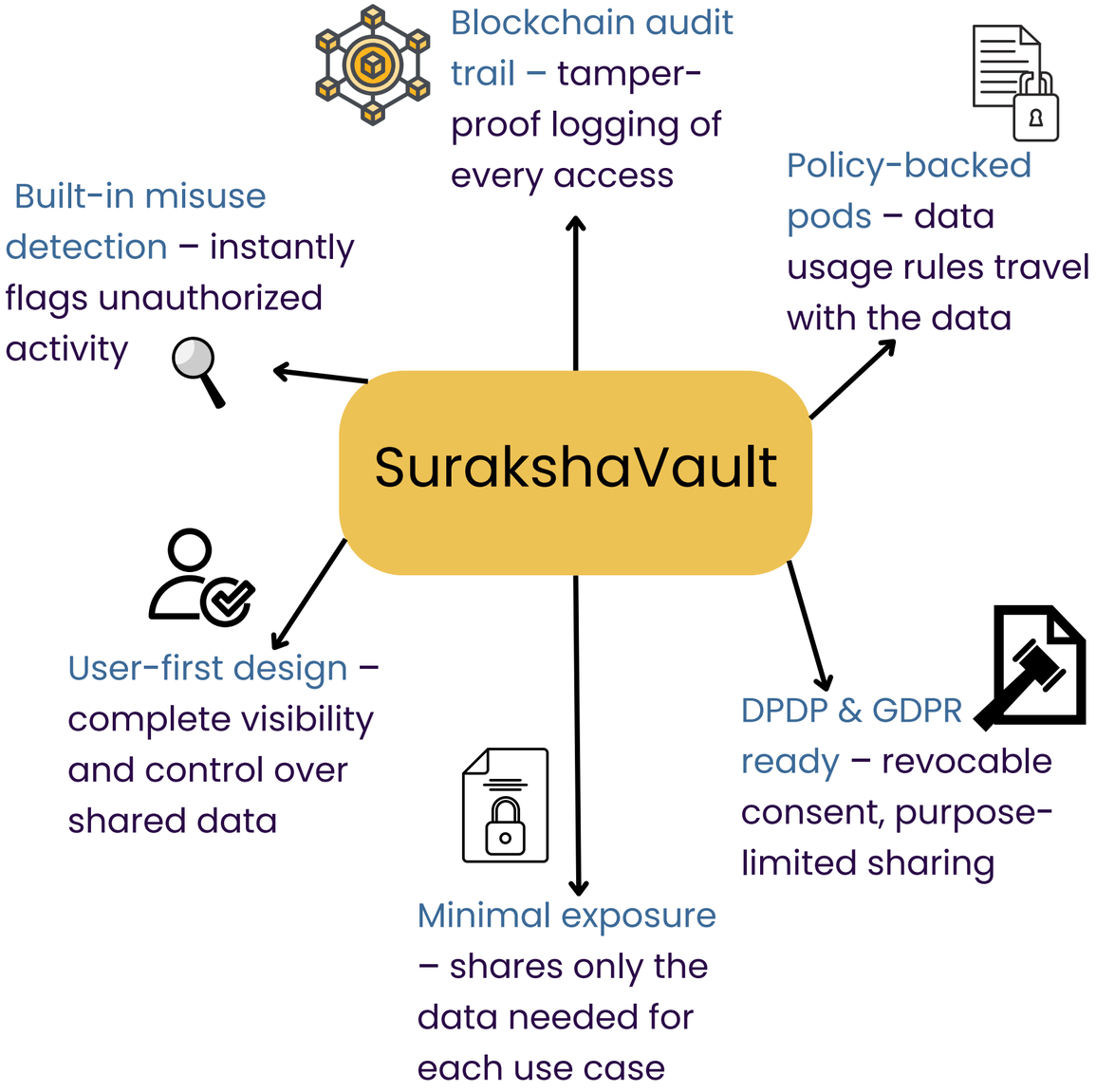
OUR SOLUTION

Suraksha Vault is a privacy-focused solution designed to enhance data sharing in fintech ecosystems while ensuring user control, compliance, and security. The core concept revolves around the creation of Data Pods that encapsulate sensitive information with embedded policies, allowing for responsible data sharing.

Step-by-Step Implementation Plan:

- Data Minimization Engine (Python + SQLite)
- Smart Policy Attachments (json - based)
- Consent Dashboard (Flask + React)
- Blockchain Audit Trail (Ethereum Testnet)
- Misuse Detection (Simple Heuristics)

Key Features at a Glance



Our Vision & Mission: Redefining Data Privacy

VISION

To create a world where individuals have complete ownership of their personal data, fostering trust and transparency in digital interactions. We envision a future where data sharing is conducted ethically and responsibly, supported by advanced technologies like blockchain, ensuring that privacy is prioritized and upheld in every transaction. Our goal is to lead the way in redefining data privacy standards, making secure and consent-driven data sharing the norm across industries.

MISSION

To empower individuals with control over their personal data by providing a secure, transparent, and user-friendly platform that enables consent-driven data sharing. We aim to enhance privacy through innovative technology, ensuring that users can share only the necessary information while maintaining full visibility and control over how their data is used.

Privacy is not a feature —
it's a fundamental right.

Privacy first

Trust and Transparency

Consent

Tech backed ethics

SECURE BY DESIGN: MINIMIZATION & SMART ACCESS CONTROL

DATA MINIMIZATION ENGINE

- Purpose-aware transformation of sensitive data
- Uses the following techniques to reduce exposure:

Technique	Example Input	Transformed Output
Masking	Credit Score: 750	7XX
Categorization	Income: ₹5,00,000	₹4L–₹6L
Aggregation	100+ Transactions	Food: ₹8K, Rent: ₹12K

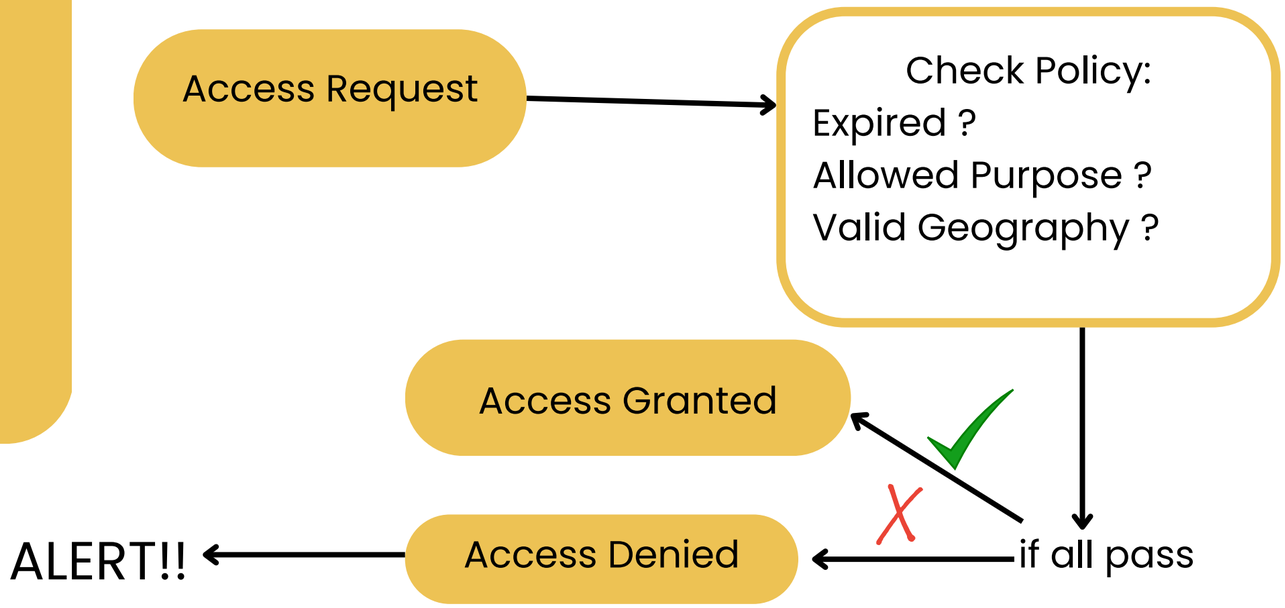
- Ensures third parties only get what they truly need
- Minimizes risk in case of misuse or leak

SMART POLICY ATTACHMENTS

Each Data Pod contains embedded machine-readable policies like:

- Validity window (e.g., expires in 30 days.)
- Allowed use cases (e.g., "loan_approval" only)
- Retention period (auto-deletes after 30 days)
- Geographic constraint (e.g., India only)
- Re-sharing disabled by default.

These policies are checked automatically before any access.



TRANSPARENT ACCESS. TOTAL CONTROL.

COMPLIANCE READY

- Fully aligned with GDPR, India's DPDP Act, and RBI guidelines
- Supports: 1. Consent Receipts for traceable approval
  2. Data Minimization by purpose
  3. Revocation Controls and Access Logs
- Demo uses synthetic (mock) data only — no real PII involved

MISUSE DETECTION

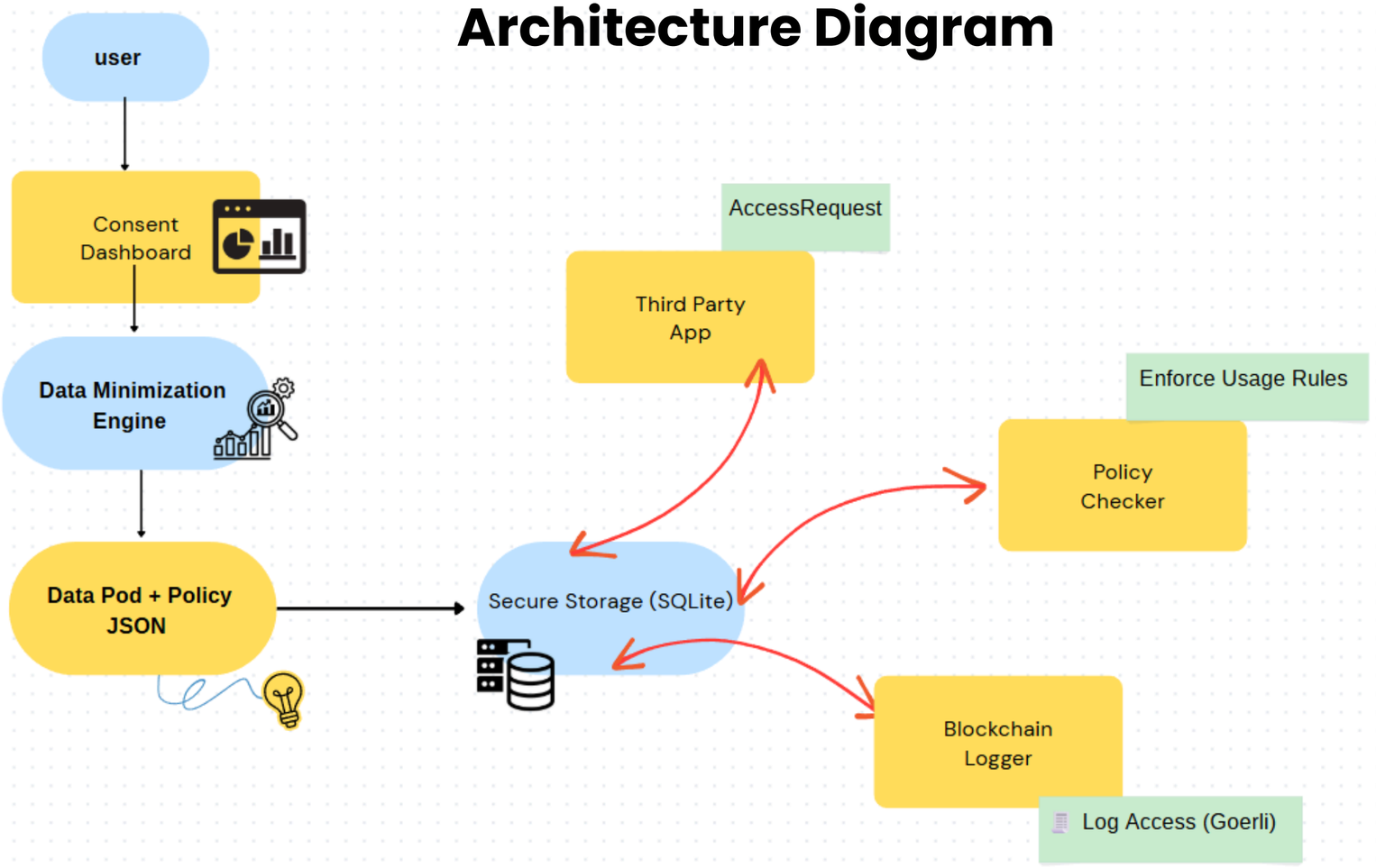
- Automatically detects policy violations like:
- Denied Access outside allowed geography
- Denied Access beyond expiry or for disallowed purpose
- Violations trigger real-time alerts and are logged to the blockchain

BLOCKCHAIN AUDIT TRAIL

- Every access request—granted or denied—is hashed and logged on the Goerli Ethereum testnet.
- Logs include who accessed, what data, when, and under which policy.
- Provides tamper-proof, independently verifiable audit history.

TECHSTACK AND ARCHITECTURE DIAGRAM

Architecture Diagram



Tech Stack

Frontend - React js + Tailwind css

Backend - Flask, Node.js

Database - SqLite

Blockchain - Ethereum Goerli testnet

Data Masking - Faker library

Data Minimization Engine - Python + SqLite

WEBSITE FLOW

User Enter Crendentials and sign up

Login

concent dashboard overview

User is directed to Consent Dashboard displaying a list of active data pods shared by the user.

User selects "Share Data" option. Submits form. Backend generates a minimized pod and returns confirmation.

Sharing Data with Third Parts

Simulating Third-Party Access

User can simulate third-party request. Backend checks policy validity and logs event.

Revoking Access

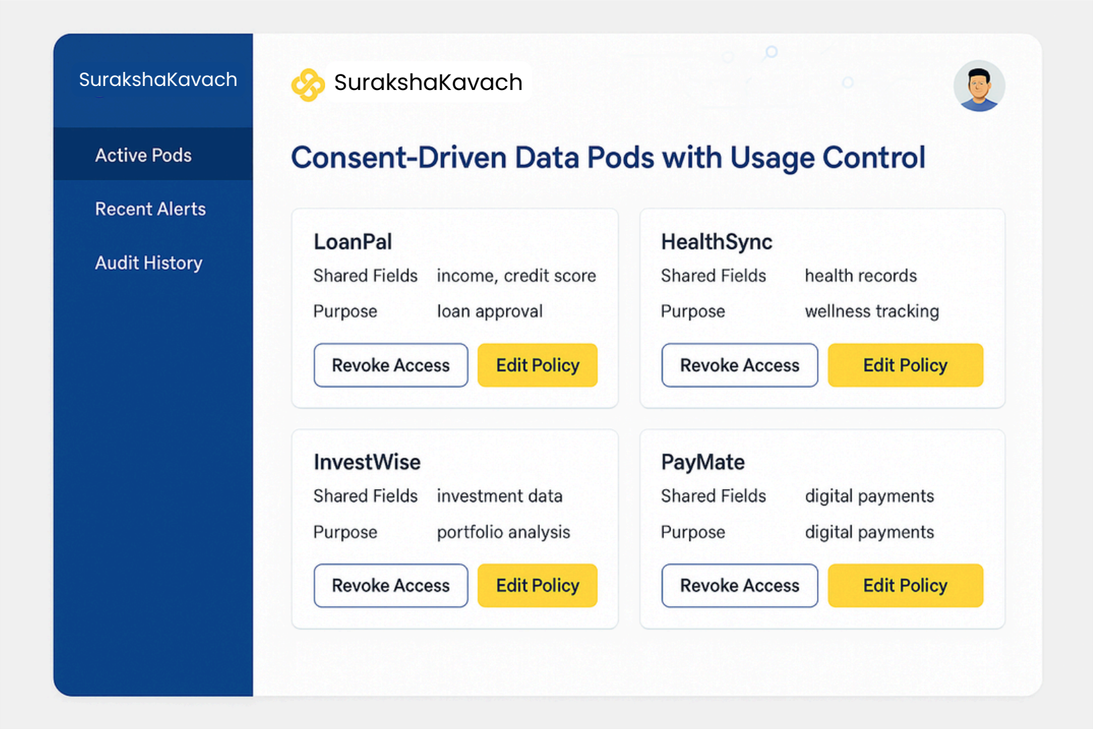
User decides to revoke access to a specific pod and clicks the button. Backend updates policy to deactivate pod.

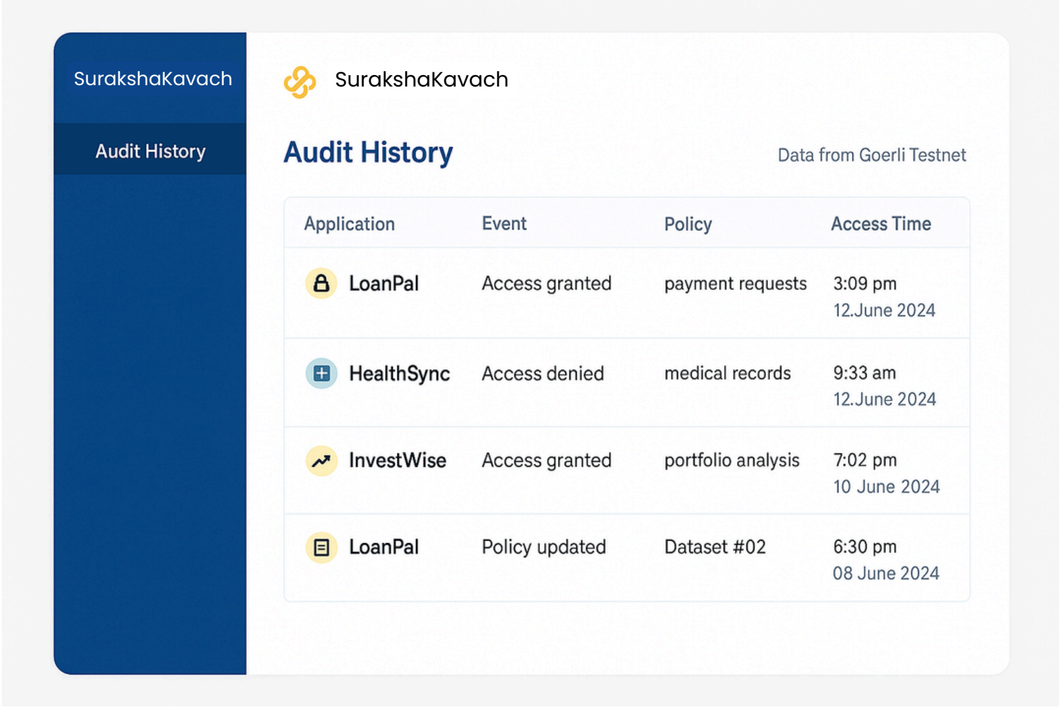
Viewing Audit Logs

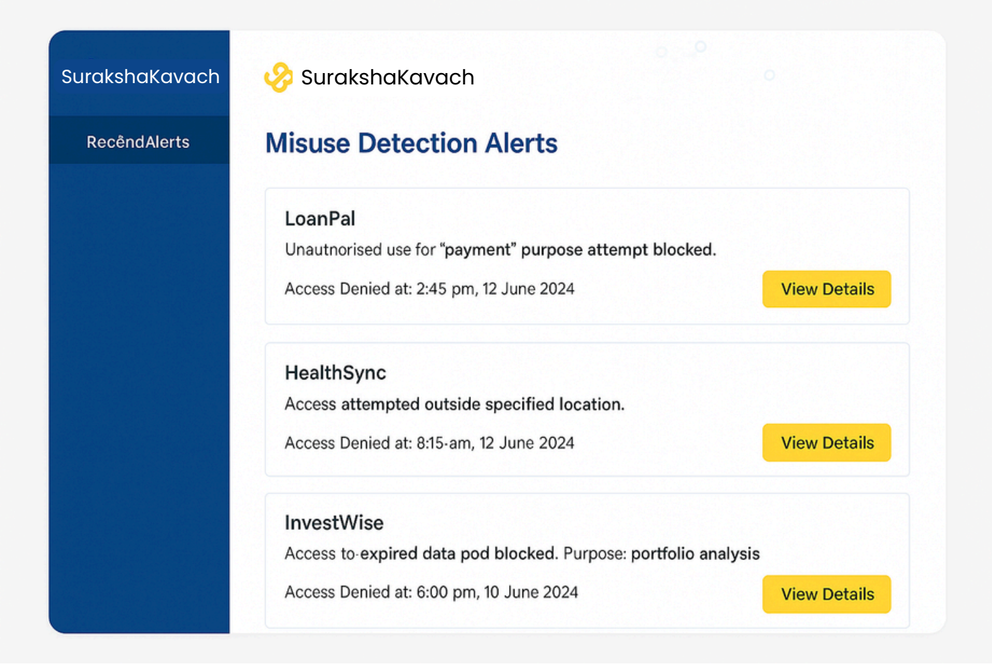
User clicks on "View Audit" for a pod. System retrieves audit logs from blockchain.

Logout

FIGMA PROTOTYPE







SOCIAL & MARKET IMPACT

Social Impact

- Empowers users with full control over their data
- Suraksha Vault gives individuals the ability to decide what data they share, with whom, for what purpose, and for how long — putting power back in the hands of the user.
- Increases public trust in fintech platforms
- By enabling transparent, revocable, and accountable data sharing, our system strengthens user confidence in digital financial services, which is essential for mass adoption.
- Supports digital rights & privacy in line with India's DPDP Act
- Our architecture aligns with India's new Digital Personal Data Protection (DPDP) Act by enforcing purpose-limited sharing, data minimization, and user consent at the core.
- Reduces risk of data misuse & breaches
- By limiting data exposure, embedding policy rules, and logging every access, Suraksha Vault significantly lowers the chances of unauthorized use, leaks, and cyber exploitation.

Market Impact

- Helps fintechs comply with GDPR, DPDP, and RBI guidelines
- Suraksha Vault supports regulatory compliance by enforcing consent, data minimization, revocation, and auditability—meeting key requirements under global and Indian data laws.
- Promotes responsible Open Banking practices
- By embedding usage policies and access controls directly into data pods, it ensures that third-party integrations follow strict, purpose-bound access—making Open Banking safer and more trusted.
- Enables new business models for secure data exchange
- With built-in consent, control, and policy enforcement, fintechs can build privacy-respecting APIs and data services, opening up revenue streams in lending, insurance, analytics, and beyond.
- Adds competitive edge through trust-focused design
- In a market where trust equals loyalty, offering transparency, revocation, and audit trails gives fintechs a clear advantage—building long-term customer relationships while minimizing risk.

Key Innovations

- Data Pods with embedded policies
- Each pod contains only essential user data and carries usage rules (purpose, expiry, geography) to enforce control even outside the platform.
- Consent Dashboard with live logs
- Users can view, monitor, and revoke third-party access in real-time—ensuring full visibility and consent-driven sharing.
- Blockchain-based audit trail
- All access events are immutably logged on the Ethereum testnet, enabling tamper-proof transparency and trust.
- Real-time misuse detection
- Requests are checked against policy rules before access. Any mismatch (e.g., expired, wrong purpose) triggers denial and alert.
- Unified privacy, usability & compliance
- Combines data minimization, policy enforcement, and auditability in a single, user-friendly and regulation-ready system.

Feasibility

- Modular architecture enables independent development
- Each component—data pods, policy engine, dashboard, and audit layer—is designed as a self-contained module, making the system easy to build, test, and scale in phases.
- Built using lightweight, beginner-friendly technologies
- Leveraging Flask (Python) for backend, React for frontend, and SQLite for quick local storage ensures rapid development without sacrificing clarity or maintainability.
- Safe and realistic simulation using mock data + testnet blockchain
- The use of synthetic data and Ethereum Goerli testnet enables safe, cost-free simulation of real-world scenarios—such as access requests, policy violations, and revocations.
- Seamless integration with fintech APIs and data ecosystems
- The solution is designed to plug into existing financial infrastructures, enabling it to fetch data, enforce consent, and log usage with minimal overhead.
- Scalable to support enterprise-grade features
- The architecture can evolve to include encryption, real-time policy editors, secure cloud storage, role-based access, and ML-based anomaly detection—without requiring a redesign.